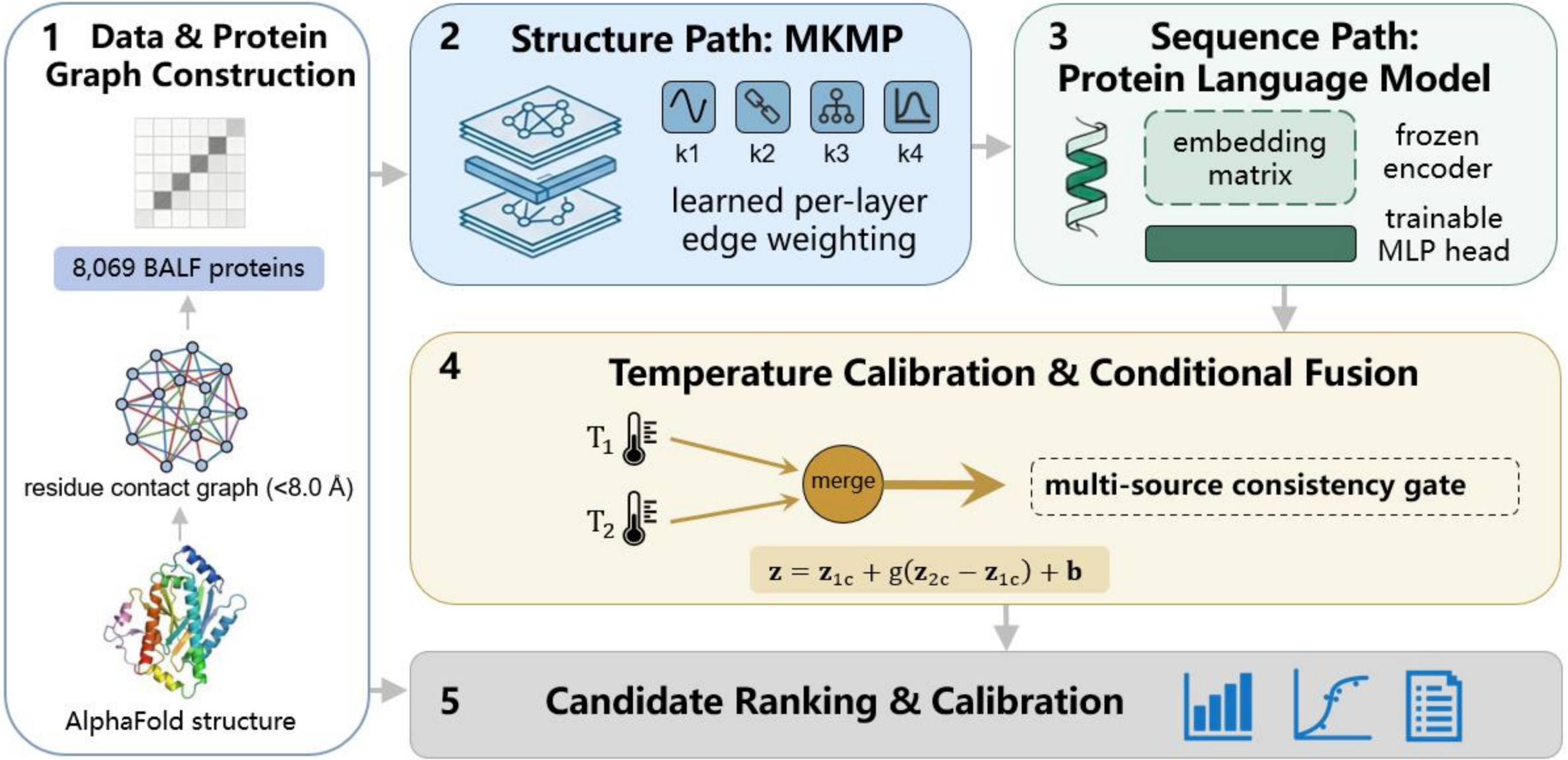

1 Data & Protein Graph Construction
8,069 BALF proteins
residue contact graph (<8.0 Å)
AlphaFold structure
2 Structure Path: MKMP
k1
k2
k3
k4
learned per-layer edge weighting
3 Sequence Path: Protein Language Model
embedding matrix
frozen encoder
trainable MLP head
4 Temperature Calibration & Conditional Fusion
T1
T2
merge
multi-source consistency gate
z = z1c + g(z2c − z1c) + b
5 Candidate Ranking & Calibration